

Program : Diploma in Textile Technology	
Course Code : 3062	Course Title: Textile Wet Processing
Semester : 3	Credits: 3
Course Category: Program core	
Periods per week: 3 (L:2, T:1, P:0)	Periods per semester: 45

Course Objectives:

- Identify the chemicals used and pre-treatments given to cotton fibres.
- To impart the knowledge of dyeing natural cellulosic & Viscose fibres using water-soluble & water-insoluble dyes.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Knowledge of chemistry		Applied chemistry	I

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the pre-treatments given to Cotton Grey fabrics	11	Understanding
CO2	Discuss various methods of bleaching textile materials & Cotton Mercerisation process. and Mercerizing	10	Understanding
CO3	Discuss the process of dyeing Cotton/Viscose fibre using water-soluble dyes.	10	Understanding
CO4	Explain the process of cotton/ viscose fibre dyeing with water insoluble dyes.	12	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the pre-treatments given to Cotton Grey fabrics		
M1.01	Memorize the chemistry of Wetting and know various chemicals used in textile wet processing	4	Remembering
M1.02	Discuss the preparation of cotton/ Viscose for dyeing	4	Understanding
M1.03	Differentiate Singeing, Desizing & Scouring	3	Understanding
Contents: Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working			
CO2	Use of various methods of bleaching textile materials & Cotton Mercerisation process.		
M2.01	Classify the bleaching process given to cotton materials	2	Understanding
M2.02	Use of continuous method of scouring and Bleaching process & compare hypochlorite and peroxide bleaching	4	Understanding
M2.03	Gain the knowledge on the process and importance of Mercerisation to implement the same in new situation.	4	Understanding
	Series Test - I	1	

Contents:

Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite.

Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching.

Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN.

CO3	Implement the process of dyeing Cotton/Viscose fibre using water-soluble dyes.		
M3.01	Classify dyes based on the method of application	2	Understanding
M3.02	Execute the application of Direct dyes, Reactive dyes on Cotton/ Viscose fibres	3	Understanding
M3.03	Execute the application of Basic dyes on Cotton material	3	Understanding
M3.04	Test the fastness properties of dyed material	2	Understanding

Contents:

Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing.

Important Natural dyes.

Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials.

Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material

CO4	Execute cotton/ viscose fibre dyeing with water insoluble dyes.		
M4.01	Execute the process of dyeing cotton and viscose materials with Water-insoluble dyes	4	Remembering
M4.02	Application of Ingrain dyes on textile materials	4	Understanding
M4.03	Test the fastness properties of dyed material	4	Understanding
	Series Test - II	1	

Contents:

Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dying and remedies.

Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material

Ingrain Dyes: Azoic Colours - Dyeing – Diazotisation - Development of Colour on Cotton

Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton.

Fastness properties of dyed material

Text /Reference:

T/R	BookTitle/Author
T1	V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan publishers Ahmedabad.
R2	S.P.Mishra- Fibre science and technology
R3	R S Prayag – Scouring, Bleaching and Mercerizing and Dyeing of cellulose fibres
R4	E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres
R5	Srivastava – Recent process in Textile Bleaching and Dyeing
R6	R RChakravarthi – A Glimpse of the chemical technology of Textile fibres
R7	Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi

Online resources:

Sl.No	Website Link
1	http://textinfo.wordpress.com/2011/10/24/classification-of-textile-fibers-according/ (accessed on 20 October 2013)
2	https://nptel.ac.in/courses/116/102/116102026/
3	https://nptel.ac.in/courses/116/102/116102016/
4	https://swayam.gov.in/
5	http://eacharya.inflibnet.ac.in/vidya-mitra/